RMSC 5001
Advanced Statistical Theory in Risk Management

Will the client subscribe a term deposit?



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1. Introduction

Telemarketing is a common sales strategy in the banking industry nowadays. Tele sales are calling their potential customers daily and seek any business opportunity from them. Most banks are trying their best to figure out the types of customers who are more willing to purchase a product (say subscribe a term deposit account) in the phone. In view of this, we are going to perform analysis on a large set of past telemarketing data to achieve the above goal. We want to identify whether any particular characteristics of customers or business indicators can predict the customer will subscribe a term deposit.

However, for small banks, it is not easy to obtain a large amount of data and the resources (e.g. labour/ customer database) for those banks are limited. Therefore, it is important to show that if a relative smaller size of data is also applicable for analysis.

2. Objectives

In view of the above, our study has two objectives:

- To investigate the nature of the customers who are likely to subscribe a term deposit in a specific bank
- To compare if 10% of sample from a full set data will provide similar results as the full set

3. Data

3.1 Background

In a direct telemarketing campaign held by a Portuguese banking institution, data were obtained to observe the relation between numbers of subscribed accounts and contact habits with the customers.

3.2 Size

We will conducts the same analysis by two dataset:

- Full Dataset: 41188 Records
- Sample Dataset (10% of the full set data): 4119 Records

3.3 Source

A Data-Driven Approach to Predict the Success of Bank Telemarketing. Decision Support Systems, Elsevier, 62:22-31, June 2014

http://achive.ics.uci.edu/ml/datasets/Bank+Marketing

[Moro et al., 2014] S. Moro, P. Cortez and P. Rita.

3.4 Variables

There are 20 variables including personal characteristics and financial data in Portugal. In our study, we will include 19 variables in our model.

Item	Variable	Description
1	age	Numeric
2	job	Categorical – Different types of job (12 Classifications)
3	marital	Categorical – Marital Status
4	education	Categorical
5	default	Categorical – Any credits in default?
6	housing	Categorical – Any housing loan?
7	loan	Categorical – Any personal loan?
8	contact	Categorical – Communication type
9	Month	Categorical – Last contact month
10	day_of_week	Categorical – Last contact week day
11	duration (*)	Numeric – Last contact duration
12	campaign	Numeric – Number of contacts with the client during this campaign
13	pdays	Numeric – Number of days passed by after the clients was contacted by previous campaign (“999” stands for “not previously contacted”)
14	previous	Numeric – No. of contacts with the client before this campaign
15	poutcome	Categorical – Outcome of previous campaign
16	emp.var.rate	Numeric – Employment Variation Rate (Quarterly Indicator)
17	cons.price.idx	Numeric – Consumer Price Index (Monthly Indicator)
18	cons.conf.idx	Numeric – Consumer Confidence Index (Monthly Indicator)
19	euribor3m	Numeric – Euribor 3-month Rate (Daily Indicator)
20	nr.employed	Numeric – No. of Employees (Quarterly Indicator)
21	Y	Binary – Target Output: Whether the client has subscribed a term deposit or not (1 – Subscribed, 0- Not Subscribed)

Note (*): The field “duration” is excluded from our scope of analysis as it is not known before the phone call is conducted. Also, the target output y is obviously known after the call is ended. Therefore, this input should be discarded if we intended to have a realistic predictive model.

4. Logistic Regression Model

4.1 Methodology

We aim to (i) find the relationship between y (i.e. whether the account is subscribed) and the 20 variables obtained and (ii) compare if the sample size matters much on the result.

Let $\pi = Pr\{ y = 1| x\}$ = Probability of the account is subscribed

$\ln [\pi/ (1-\pi)] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n$

In the logistic regression, we assume that the log-odd ratio (i.e. $\ln [\pi/ (1-\pi)]$)of success probability is a linear function of different variables.

Sine the number of observations is huge, we first divide the whole dataset into two exclusive parts: training dataset (d1) and testing dataset (d2). 90% observations are randomly selected as training dataset to build the logistic regression model while the rest as testing dataset to assess the accuracy of the model.

In order to have a good estimate of β, the R built-in glm() function is used whereas the last option “binomial” is the link function of logistic regression and the output is stored in lreg. By first run, the maximum likelihood estimation is large due to many outliers exist in the dataset.

Therefore, carrying out backward elimination to eliminate variables with large p-values one by one would be essential. To do this, the stepwise function step() is used. An Akaike Information Criterion (AIC) would be computed in each step as a measure of how well the model fit the data. The stepwise function shall stop when it does not perceive that choosing a smaller AIC would yield much additional improvement in the model.

When the regression model with most variables are significant is obtained, function predict() is implemented and assisted in building the classification table and calculating the misclassification rate. The first objective is then achieved.
In addition, the above logistic regression will be implemented twice on both full set and 10% of the dataset. We would like to observe the difference in the significant variables and the machine time for running the programme.

4.2 Results & Findings

Regression

Both regression models on two datasets are generated with many insignificant variables included. However, further removing all the insignificant ones may affect the regression results. Hence, instead of elimination, we shall extract the significant variables with "*", "**", "***" to compare and correspond to our objective on finding out factors that are highly related to the account subscription.

Full dataset

Significant Coefficients	Estimate	Pr(>|z|)
cons.price.idx	2.000E+00	***
monthmar	1.447E+00	***
poutcomesuccess	8.524E-01	***
poutcomenonexistent	5.149E-01	***
monthaug	4.428E-01	***
euribor3m	2.758E-01	*
jobstudent	1.958E-01	*
jobretired	1.674E-01	*
cons.conf.idx	2.628E-02	***
pdays	-9.573E-04	***
campaign	-4.035E-02	***
jobservices	-1.562E-01	*
jobblue-collar	-2.135E-01	***
day_of_weekmon	-2.190E-01	***
defaultunknown	-2.430E-01	***
monthmay	-4.246E-01	***
monthnov	-4.899E-01	***
monthjun	-6.575E-01	***
contacttelephone	-7.318E-01	***
emp.var.rate	-1.494E+00	***
(Intercept)	-2.163E+02	***

10% dataset

Significant Coefficients	Estimate	Pr(>|z|)
monthmar	1.705E+00	***
cons.price.idx	1.270E+00	***
poutcomenoexistent	4.321E-01	*
cons.conf.idx	4.919E-02	**
campaign	-6.758E-02	*
monthmay	-4.722E-01	*
emp.var.rate	-7.151E-01	***
contacttelephone	-8.628E-01	***
(Intercept)	-1.184E+02	***

The more positive the coefficients are, the higher the probability of success can be achieved, vice versa.

In both sets of data, we observe the variables with significance and coefficient order are similar.

"month", "cons.price.idx", "poutcome", "cons.conf.idx", "campaign", ëmp.var.rate" and "contact" are highly related variables in both dataset. In the full dataset, "job", "date_of_work", "pdays", "default" are also included in the regression model.

Misclassification Rate

Full dataset	10% dataset
Training Data	
pred 0 1 0 32394 3224 1 488 963 **Misclassification Rate:** 10.013760%	pred 0 1 0 3247 312 1 50 98 **Misclassification Rate:** 9.765309%
Testing Data	
pred 0 1 0 3615 341 1 51 112 **Misclassification Rate**: 9.516873%	pred 0 1 0 363 33 1 8 8 **Misclassification Rate:** 9.951456%

Training data in the full set of data shows a higher classification rate in the full set of data while testing data show a higher misclassification rate in the 10% dataset. However, the difference is less than 0.5% which is insignificance.

Testing error rate is more realistic to reflect the true performance of the model and hence, the model with full set of data is performing better than the other one. Nevertheless, the difference is again less than 0.5% and we can assume the error rate on the 40,000up data versus the 4,000up data is similar in this regression model.

Hence, both datasets show the performance of the model is good to analyse the relative variables in subscribing a term deposit account.

Machine Time

Full dataset	10% dataset
Machine Time	
2.61 mins	56.59 secs

Running time for implementing the full set of data is much longer than the 10% of it, but still within an acceptable mean of time. The regression model can be said as fitting both large or smaller sample size of data.

5. Classification Tree

5.1 Methodology

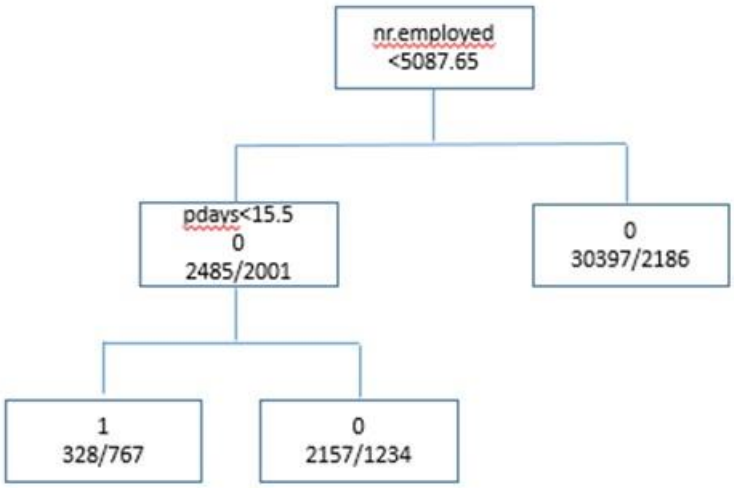
Classification Tree based on the binary splitting of variables, one at a time. It is constructed such that each terminal node is as “pure” as possible. Therefore, in each terminal node, most of the observations are belongs to the same group. Based on the Classification Tree, a set of simple classification rules can be easily obtained. R has build-in function to apply the Classification Tree.

By using R, we want to find out which variables can significantly identify the client subscribed a term deposit. At the beginning, we put 19 variables into R program and divide training dataset and testing dataset. We randomly select a subset from the dataset as the training dataset (i.e. 90% data) and build a Classification Tree model. The rest of the data (i.e. 10% data) will form the testing dataset to test the model.

5.2 Results & Findings

Full Dataset

The dataset is split into two classes (0 or 1) with 1 indicates that client subscribed a term deposit while 0 indicates that client did not subscribe a term deposit. The classification tree is tabled as bellows.



We got total 3 nodes and the classification rules are summarized as below.

Node1: If nr_employed >=5087.65 then Result = 0 (30397/2186)
Node2: If (nr_employed <5087.65) and (pdays >= 15.5) then Result = 0 (2157/1234)
Node3: If (nr_employed <5087.65) and (pdays < 15.5) then Result = 1 (328/767)

The result shows that client subscribed a term deposit when nr_employed < 5087.65 and pdays < 15.5.

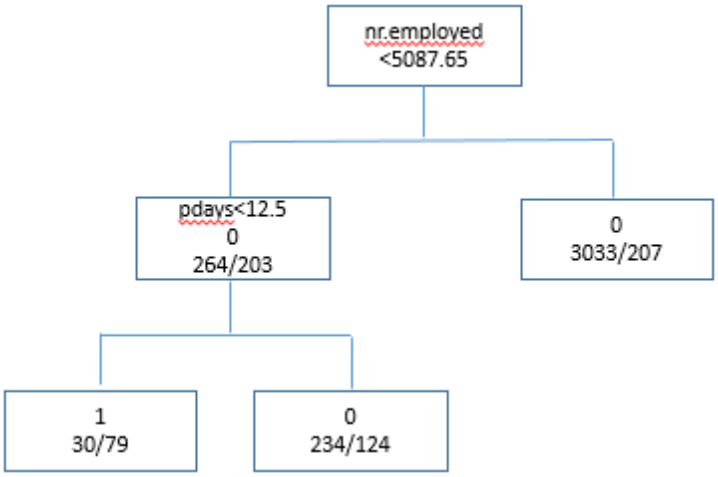
We noted that the support rate, confidence rate and capture rate of node 1 are 87.4%, 93.6% and 92.0% respectively. Therefore, we can be much sure that if nr.employed is larger than or equal to 5087.65, client did

not subscribed a term deposit.

10% Dataset

Apart from using the full dataset in Classification Tree, we also try to use 10% of the whole dataset for model building. We find that the classification rule is similar as using the whole dataset. Besides, the predictive power of the model (in terms of misclassification rate) even perform better.

The classification tree is tabled as follows.



Again, we got total 3 nodes and the classification rules are summarized as below.

Node1: If nr_employed >=5087.65 then Result = 0 (3033/207)
Node2: If (nr_employed <5087.65) and (pdays >= 12.5) then Result = 0 (234/124)
Node3: If (nr_employed <5087.65) and (pdays < 12.5) then Result = 1 (30/79)

We can see that client subscribed a term deposit when nr_employed < 5087.65 and pdays < 12.5. The parameters are the same as using the whole dataset. The only difference is the threshold of pdays changes from 15.5 (whole dataset) to 12.5 (10% dataset).

The support rate, confidence rate and capture rate of node 1 in 10% dataset are 87.9%, 93.3% and 92.4% respectively. The result aligns the whole dataset.

We also compared the misclassification rate between training dataset and testing dataset. The result shows that the model is stable and performs well.

Misclassification Rate

In order to test the stability of the model result, we compared the misclassification rate between training dataset and testing dataset. The result shows that the model is stable and performs well.

Full dataset	10% dataset
Training Data	
col 0 1 1 32554 3420 2 328 767 **Misclassification Rate**: 10.11087%	col 0 1 1 3267 331 2 30 79 **Misclassification Rate:** 9.738333%
Testing Data	
col2 0 1 1 3640 367 2 26 86 **Misclassification Rate**: 9.541151%	col2 0 1 1 366 32 2 5 9 **Misclassification Rate:** 8.980583%

There is an interesting finding that the threshold of nr_employed is 5087.65 under the model of using full data set or just 10% of the dataset and the predictive power (in terms of misclassification rate) of 10% dataset is better.

Machine Time

Full dataset	10% dataset
Machine Time	
11.54 secs	1.63 secs

Running time for using the full dataset is much longer than 10% dataset. However, the result is not significant better. Therefore, we propose to adopt 10% dataset instead of full dataset in future study.

6. Artificial Neural Network

6.1 Methodology

Apart from Logistic Regression and Classification Tree, we apply Artificial Neural Network in our study. Artificial Neural Network is to mimic the functions and mechanism of our brain.

In order to apply Artificial Neural Network we need to consider (1) size of the hidden layer, (2) number of hidden layer, (3) number of inputs, (4) size of the training sample and testing sample and (5) type of output (logistic or linear).

The optimal hidden layer size for the Artificial Neural Network has to be found out by doing a number of trials. Therefore, we design a new ANN function by forming a loop to determine the best hidden layer size and then save the best training result among different hidden layer size. We choose hidden layer maxize = 18 because the number of weights exceed the default limit of 1024. It is possible to exceed the default limit but this is undesirable because doing so only increases complexity.

The model is using back-propagation algorithm and find the appropriate weight w such that E = (Y-V)'(Y-V) is minimized, where V is the prediction from Artificial Neural Network. We build an improved version of the model to test the size from 1 to 10 and select the best variable from 1 to 18.

6.2 Results & Findings

In order to test the stability of the model, we randomly select 90% of data as training dataset and the rest of 10% data as testing dataset and the result (in terms of misclassification rate) are summarized as below.

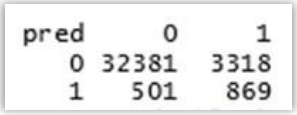
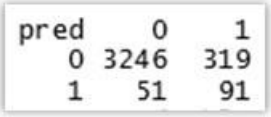
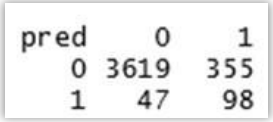
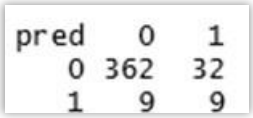
Full dataset

Again, 1 indicates that client subscribed a term deposit while 0 indicates that client did not subscribe a term deposit. The model finally stop at weights 973 with the model 52-18-1.

10% dataset

If we use 10% dataset, the model stop at weights 757 with the model 52-14-1. The total layer required is similar for both dataset.

Misclassification Rate

Full dataset	10% dataset
Training Data	
 **Misclassification Rate**: 10.30241%	 **Misclassification Rate:** 9.981117%
Testing Data	
 **Misclassification Rate**: 9.75965%	 **Misclassification Rate:** 9.951456%

By comparing the misclassification rate of both dataset, we see that the overall performance are similar and the predictive power of them are good.

Machine Time

Full dataset	10% dataset
Machine Time	
2.99 mins	20.8 secs

Running time for using the full dataset is much longer than 10% dataset. However, the result is not significant better. Therefore, we propose to adopt 10% dataset instead of full dataset in future study.

We got an interesting finding which are deviated from our original thought. Originally, we thought that applying Artificial Neural Network can largely enhance predictive power of the model and help us select more significant variables to predict whether client will subscribe a term deposit. However, the result shows that the misclassification rate between Classification Tree and Artificial Neural Network are similar and there is no significant different between them. Therefore, in the consideration of large number of variables and coefficients involved in Artificial Neural Network, we suggest to use Classification Tree to predict whether client will subscribe a term deposit.

7. Conclusion

According to the analysis results from the above 3 models, we can conclude that a customer who is more likely to subscribe a term deposit probably have the following characteristics:

- Last Contact Month: March
- High Consumer Price Index (Monthly Indicator)
- High Consumer Confidence Index (Monthly Indicator)
- Not participating in any marketing campaign before
- No. of Employees (Quarterly Indicator): Less Than 5087.65
- Number of days passed by after the clients was contacted by previous campaign: Less Than 12.5

The low misclassification rates from the outputs of our model also implies that the results are reliable.

Also, when the full set data and the 10% sample dataset are applied to the 3 models, similar error rates and results were obtained. We can conclude that the results from a smaller dataset can also provide a referenceable result for small banks to target their customers.

8. Appendix

Please refer to separate document for details.